

Unit 6.5: Cubic & Cube Root Functions Translations Practice

Name: _____

Period: _____

Process: (1) Find h and k (2) Plot center point (h, k) (3) Shift anchor points (4) Draw the curve

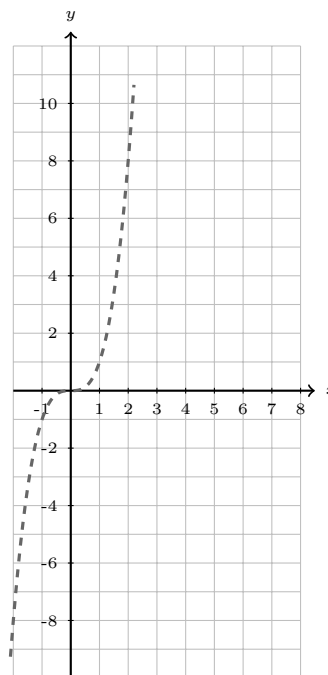
1. $f(x) = (x - 3)^3 + 2$

Verbal Desc: _____

$h =$ _____ $k =$ _____

Center point: (,)

Parent	Shifted
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)



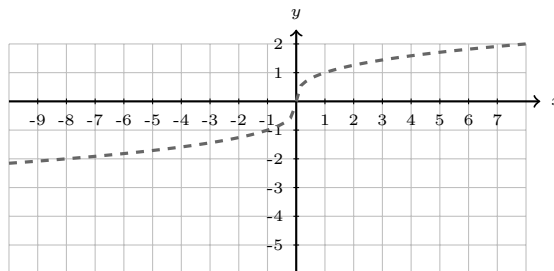
2. $f(x) = \sqrt[3]{x+1} - 3$

Verbal Desc: _____

$h =$ _____ $k =$ _____

Center point: (,)

Parent	Shifted
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)



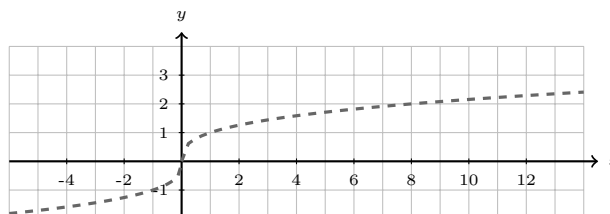
3. $f(x) = \sqrt[3]{x-4} + 1$

Verbal Desc: _____

$h =$ _____ $k =$ _____

Center point: (,)

Parent	Shifted
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)



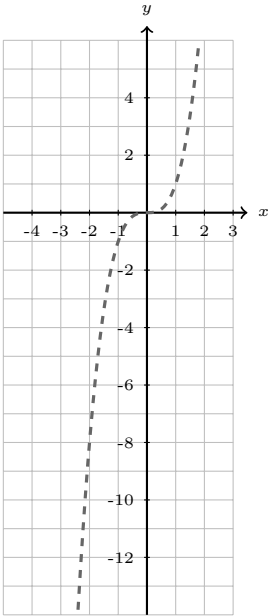
4. $f(x) = (x + 2)^3 - 4$

$h = \rule{1cm}{0.4pt}$ $k = \rule{1cm}{0.4pt}$

Center point: (,)

Parent	Shifted
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)

Verbal Desc: _____



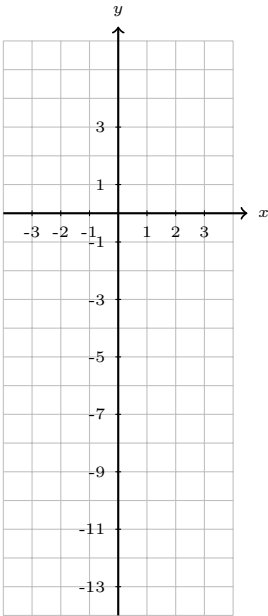
5. $f(x) = x^3 - 5$

$h = \rule{1cm}{0.4pt}$ $k = \rule{1cm}{0.4pt}$

Center point: (,)

Parent	Shifted
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)

Verbal Desc: _____



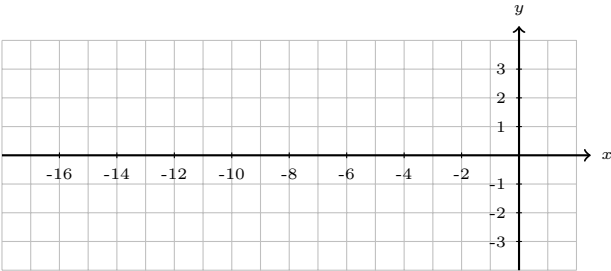
6. $f(x) = \sqrt[3]{x + 8} + 1$

$h = \rule{1cm}{0.4pt}$ $k = \rule{1cm}{0.4pt}$

Center point: (,)

Parent	Shifted
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)

Verbal Desc: _____



Translations Practice (continued)

Name: _____

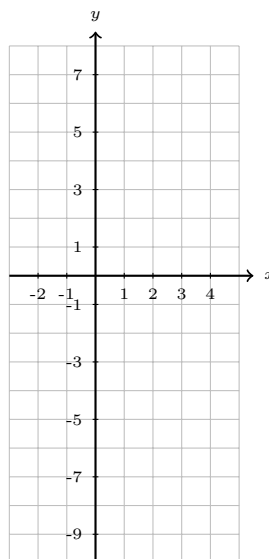
7. $f(x) = (x - 1)^3 - 1$

Verbal Desc: _____

$h =$ _____ $k =$ _____

Center point: (,)

Parent	Shifted
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)



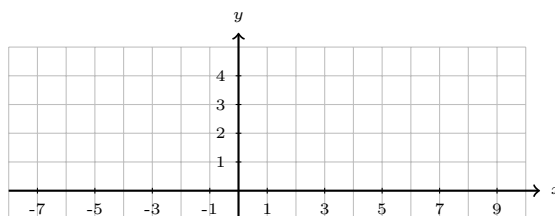
8. $f(x) = \sqrt[3]{x - 1} + 2$

Verbal Desc: _____

$h =$ _____ $k =$ _____

Center point: (,)

Parent	Shifted
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)



Summary Questions:

9. Write the equation of a cubic function with center point at $(-3, 5)$.

10. Write the equation of a cube root function with center point at $(4, -2)$.

11. A student says that $f(x) = (x + 5)^3$ shifts the graph *right* 5 units. Are they correct? Explain.